

AI Support Ticket Management System

Complete Project Architecture • Roles • M1–M3 Flow • AI Agents • Dashboards • RAG • Test Data • Testing Scenarios

Project Goal

Build an intelligent support platform where customers raise issues, AI understands and analyzes tickets, retrieves relevant knowledge, generates suggested resolutions, automatically resolves simple cases, and escalates complex cases to human support agents.

Document Scope

Area	Coverage
M1	Core ticket lifecycle, users, tickets, assignment and communication
M2	AI classification, priority prediction, sentiment and similar-ticket intelligence
M3	RAG knowledge retrieval, AI resolution, confidence scoring and escalation
Users	Customer, Support Agent, Support Manager, Admin
AI Layer	Orchestrator, Classification, Priority, Sentiment, Retrieval, Resolution and Escalation Agents
Outcome	Auto-resolution where safe + human support where necessary

1. System Users and Roles

1. Customer / End User

Creates support tickets, provides issue details, uploads attachments, receives AI or human responses, tracks progress and confirms resolution.

2. Support Agent

Handles tickets escalated to humans, reviews AI analysis and suggestions, communicates with customers and resolves issues.

3. Support Manager

Monitors queues, SLA, escalations, workload and agent performance. Can assign or reassign tickets.

4. Admin

Controls users, roles, categories, SLA policies, automation rules, knowledge base governance, AI monitoring and audit logs.

Role Connection

Customer creates a ticket → AI analyzes the issue → simple/high-confidence issues receive an automated response → complex or risky cases are escalated to a Support Agent → Support Manager monitors operations and SLA → Admin manages the entire platform.

2. M1 → M2 → M3: Connected Milestone Architecture

M1 — Core Ticket Management

M1 is the operational foundation. It creates tickets, stores customer data, maintains status, supports assignment and provides the ticket lifecycle.

Customer → Create Ticket → Validation → Ticket ID → Status OPEN → Queue / Assignment

M2 — AI Ticket Intelligence

M2 consumes the ticket created by M1. AI/NLP analyzes the text to identify category, urgency, sentiment and similarity to previous issues.

M1 Ticket → AI Classification → Priority Prediction → Sentiment Analysis → Similar Ticket Detection

M3 — AI Resolution & Automation

M3 uses M2 analysis and searches trusted knowledge. RAG retrieves relevant documents, then the AI generates a grounded suggested answer. A confidence policy determines auto-response or human escalation.

AI Analysis → Knowledge Retrieval / RAG → Resolution Generation → Confidence Check → Auto Resolve OR Escalate

Important Connection: M1 creates the ticket and lifecycle data. M2 understands the ticket. M3 tries to resolve the ticket intelligently. All milestones form one connected system.

3. Complete End-to-End Project Flow

Step 1 — Customer Login

Customer logs into the portal.

Step 2 — Create Ticket

Customer enters subject, description, optional category and attachment.

Step 3 — Validation & Ticket Creation

Backend validates the request and generates a Ticket ID.

Step 4 — M1 Lifecycle Starts

Ticket is stored with status OPEN and timestamp information.

Step 5 — AI Orchestrator Starts

The Orchestrator receives the new ticket and coordinates required AI agents.

Step 6 — Classification

AI predicts category and subcategory.

Step 7 — Priority & Severity

AI estimates Low, Medium, High or Critical priority.

Step 8 — Sentiment Analysis

AI identifies customer tone such as Positive, Neutral, Negative or Highly Frustrated.

Step 9 — Similar Ticket Search

The system searches historical resolved tickets for similar issues.

Step 10 — Knowledge Retrieval

Relevant FAQ, manuals, policies and technical documents are retrieved.

Step 11 — AI Resolution Generation

The Resolution Agent uses trusted context to generate a suggested response.

Step 12 — Confidence & Safety Check

The system evaluates whether the answer is sufficiently reliable for automated handling.

Step 13 — Decision

High-confidence/simple issue → auto-response. Low-confidence/complex/high-risk issue → human escalation.

Step 14 — Human Support

Support Agent reviews the full AI analysis and responds if escalation occurs.

Step 15 — Resolution

Ticket is marked RESOLVED after solution delivery.

Step 16 — Closure

Customer confirms resolution or the system closes according to policy.

Master Process Flow

CUSTOMER → CREATE TICKET → M1 TICKET LIFECYCLE → M2 AI UNDERSTANDING → M3 RAG + AI RESOLUTION → CONFIDENCE CHECK → AUTO RESPONSE OR HUMAN ESCALATION → RESOLUTION → CUSTOMER CONFIRMATION → CLOSED

4. AI Intelligence: What Does AI Actually Do?

AI Ticket Classification

Reads subject and description and predicts category/subcategory. Example: 'Unable to login' → Account → Login Issue.

AI Priority Prediction

Estimates urgency using issue type, language, impact and historical patterns.

Sentiment Analysis

Detects customer tone and frustration level to support prioritization.

Similar Ticket Detection

Uses embeddings/vector similarity to find related historical tickets and successful resolutions.

RAG Knowledge Retrieval

Retrieves relevant knowledge from trusted FAQs, manuals, policies and support documents before answer generation.

AI Resolution Generation

Creates a contextual suggested response based on ticket analysis and retrieved evidence.

Confidence Scoring

Evaluates confidence/relevance and determines whether automation is appropriate.

Escalation Intelligence

Routes complex, low-confidence, critical or SLA-sensitive tickets to the right human queue.

5. AI Agents and Responsibilities

AI Agent	Input	Responsibility	Output
AI Orchestrator	New ticket	Coordinates workflow and agent calls	Combined analysis
Classification Agent	Subject + description	Predicts category/subcategory	Classification
Priority Agent	Ticket context	Predicts urgency/severity	Priority level
Sentiment Agent	Customer text	Detects tone/frustration	Sentiment score
Similarity Agent	Ticket embedding	Finds related historical tickets	Similar cases
Knowledge Retrieval Agent	Question + context	Searches KB/vector database	Relevant documents
Resolution Agent	Ticket + retrieved context	Generates suggested answer	AI resolution
Escalation Agent	Confidence + priority + SLA	Chooses automation or human routing	Decision + queue

Agent Connection Flow

AI ORCHESTRATOR → CLASSIFICATION + PRIORITY + SENTIMENT + SIMILARITY → KNOWLEDGE RETRIEVAL → RESOLUTION AGENT → CONFIDENCE CHECK → ESCALATION AGENT → AUTO RESPONSE / SUPPORT AGENT

6. RAG Knowledge Flow

RAG (Retrieval-Augmented Generation) helps prevent unsupported AI answers. The system retrieves relevant trusted knowledge first and then generates an answer using that context.

Knowledge Sources

FAQ • User Manuals • Product Documentation • Policies • Troubleshooting Guides • Previous Verified Resolutions • Internal Support Articles

RAG Flow

CUSTOMER TICKET → CREATE EMBEDDING / SEARCH QUERY → VECTOR DATABASE / KNOWLEDGE SEARCH → RETRIEVE RELEVANT DOCUMENTS → PASS CONTEXT TO LLM → GENERATE GROUNDED RESPONSE → CONFIDENCE CHECK

Example

Ticket: 'I cannot reset my password.'
Retrieved Article: Password Reset Troubleshooting Guide
AI Response: Provides the verified reset steps and escalation instructions if the process fails.

7. Dashboard Architecture

Recommended portals: Customer Portal, Support Agent Portal, Support Manager Portal and Admin Portal.

A. Customer Portal

Side Navigation: Dashboard • Create Ticket • My Tickets • Ticket Details • Knowledge Base • Notifications • Profile

Main Content: Cards: Total Tickets • Open • In Progress • Resolved • Recent Tickets • Notifications

B. Support Agent Portal

Side Navigation: Dashboard • My Tickets • Assigned Tickets • AI Suggestions • Knowledge Base • Escalated Tickets • SLA Alerts • Customer History • Notifications • Profile

Main Content: Cards: Assigned Tickets • High Priority • SLA Risk • Resolved Today • AI-Assisted Tickets

C. Support Manager Portal

Side Navigation: Dashboard • All Tickets • Ticket Queue • Agent Assignment • Escalations • SLA Management • Agent Performance • AI Performance • Reports • Notifications • Profile

Main Content: Cards: Open Tickets • High Priority • SLA Breaches • Escalations • Average Resolution Time

D. Admin Portal

Side Navigation: Dashboard • Users • Customers • Support Agents • Support Managers • Roles & Permissions • Categories • Subcategories • Priorities • Status Configuration • AI Agent Monitor • AI Confidence Analytics • Knowledge Base • Training Dataset • SLA Policies • Escalation Rules • Automation Rules • Assignment Rules • Reports • Settings • Audit Logs

Main Content: Cards: Total Tickets • Open • Resolved • Escalated • AI Resolved • Human Resolved • SLA Breaches • AI Performance

8. Dashboard-to-Process Connection

Customer Flow

Create Ticket → View AI/Human Response → Track Status → Reply if Required → Confirm Resolution

Support Agent Flow

Escalated Ticket → View Customer Context + AI Analysis + Suggested Answer → Investigate → Reply → Resolve

Support Manager Flow

Monitor Queue → Check SLA → Assign/Reassign → Handle Escalations → Monitor Agent & AI Performance

Admin Flow

Manage Users + Categories + SLA + Knowledge Base + Automation + AI Monitoring + Audit + Reports

Complete Portal Connection

CUSTOMER PORTAL → TICKET SERVICE → AI ORCHESTRATOR → AI AGENTS + RAG → DECISION ENGINE → AUTO RESPONSE OR SUPPORT AGENT PORTAL → CUSTOMER → MANAGER MONITORING → ADMIN GOVERNANCE

9. Complete Example: Login Issue

Customer Input

Field	Value
Ticket ID	TKT-1001
Subject	Unable to Login
Description	I changed my password yesterday and now cannot access my account.
Attachment	login_error.png
Initial Status	OPEN

AI Processing Result

AI Component	Example Result
Classification	Account → Login Issue
Classification Confidence	96%
Priority	MEDIUM
Sentiment	NEGATIVE
Similar Tickets Found	TKT-890, TKT-722
Knowledge Article	Password Reset Troubleshooting
AI Answer Confidence	94%
Decision	AUTO RESPONSE
Final Status	RESOLVED / WAITING CUSTOMER CONFIRMATION

Example Flow

Unable to Login → Classification Agent identifies Login Issue → Priority Agent assigns Medium → Retrieval Agent finds password article → Resolution Agent generates steps → Confidence 94% → Automated response → Customer confirms → Ticket Closed

10. Escalation Example: Duplicate Payment

Field	Value
Ticket ID	TKT-1002
Subject	Payment deducted twice
AI Category	Billing → Duplicate Payment
Priority	HIGH
Sentiment	HIGHLY NEGATIVE
AI Confidence	62%
Decision	ESCALATE
Assigned Queue	Billing Support
Final Handler	Human Support Agent

Escalation Flow

Duplicate Payment Ticket → Classification → HIGH Priority → Similar Cases Search → Knowledge Retrieved → Confidence only 62% → Escalation Agent routes to Billing Support → Agent investigates transaction → Refund/Correction → Customer notified → Closed

11. Sample Test Data

Ticket ID	Subject	Category	Priority	Sentiment	Expected Route
TKT-1001	Unable to login	Account	Medium	Negative	AI Auto Response
TKT-1002	Payment deducted twice	Billing	High	Negative	Human Escalation
TKT-1003	How to change email?	Account	Low	Neutral	AI Auto Response
TKT-1004	Application crashes	Technical	High	Negative	Technical Agent
TKT-1005	Need invoice copy	Billing	Low	Neutral	AI Auto Response
TKT-1006	Data missing from dashboard	Technical	Critical	Negative	Immediate Escalation
TKT-1007	Feature request	Product	Low	Positive	Product Queue

12. Core Testing Scenarios

#	Test Scenario	Expected Result
1	Customer creates valid ticket	Ticket ID generated and status OPEN
2	Subject/description missing	Validation error
3	Classification service works	Category and subcategory stored
4	Priority prediction works	Priority assigned
5	Sentiment analysis works	Sentiment score stored
6	Similar ticket search works	Relevant historical tickets returned
7	Knowledge retrieval finds article	Relevant context passed to AI
8	AI confidence above threshold	Automated response allowed
9	AI confidence below threshold	Escalate to human agent
10	Critical priority detected	Immediate alert and priority routing
11	SLA threshold breached	Manager/escalation notification
12	Agent resolves ticket	Status changes to RESOLVED
13	Customer confirms resolution	Ticket changes to CLOSED
14	Unauthorized access attempt	Access denied
15	Knowledge service unavailable	Fallback to human support queue

13. SLA and Escalation Flow

SLA ensures that tickets are handled within the expected response and resolution time.

Priority	Example Response Target	Example Resolution Target	Escalation
Low	24 Hours	72 Hours	Normal queue
Medium	8 Hours	24 Hours	Agent alert
High	2 Hours	8 Hours	Manager visibility
Critical	30 Minutes	4 Hours	Immediate escalation

SLA Flow

TICKET CREATED → PRIORITY ASSIGNED → SLA TIMER STARTS → RESPONSE TARGET CHECK → RESOLUTION TARGET CHECK → WARNING → BREACH ALERT → MANAGER ESCALATION

14. Ticket Status Flow

Recommended Status Lifecycle

DRAFT → OPEN → AI_ANALYZING → AI_RESPONDED → IN_PROGRESS → RESOLVED → CLOSED

Human Escalation Lifecycle

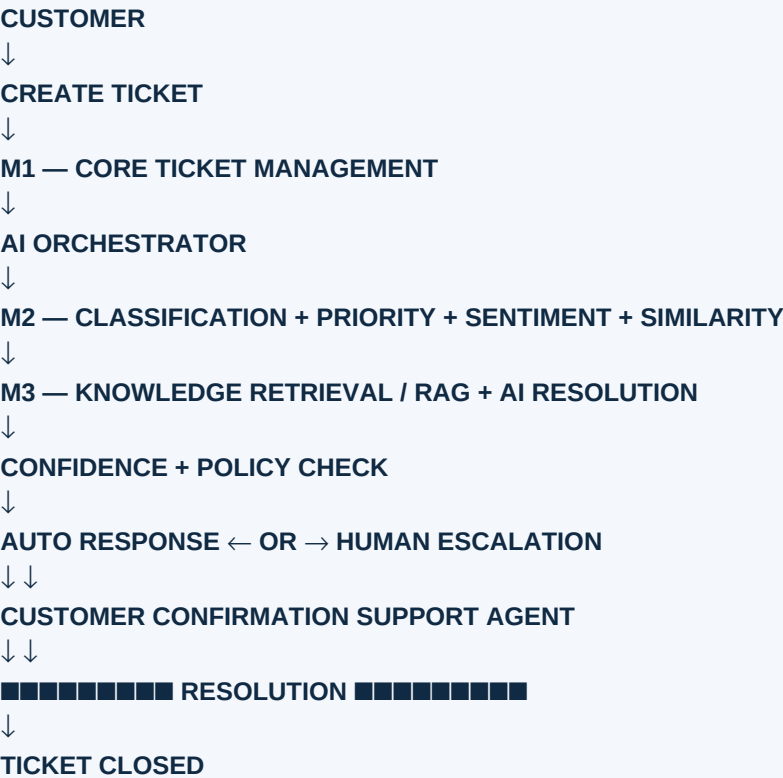
OPEN → AI_ANALYZING → ESCALATED → ASSIGNED → IN_PROGRESS → WAITING_FOR_CUSTOMER → RESOLVED → CLOSED

Optional Reopen Flow

RESOLVED → CUSTOMER REPLIES / ISSUE CONTINUES → REOPENED → IN_PROGRESS → RESOLVED → CLOSED

15. Final Complete System Architecture

Full Architecture Flow



System Governance

Support Manager monitors SLA, queues, workload and escalations. Admin controls users, roles, categories, knowledge base, automation, AI performance, reports and audit logs.

Final Project Understanding

The project is one connected AI-powered support ecosystem. M1 manages the support transaction, M2 understands the customer's issue, and M3 retrieves knowledge and generates intelligent resolutions. Automation is used only when confidence and business rules allow it; otherwise, the complete AI context is handed to a human agent for faster and more informed resolution.